

Computer Science & IT

Database Management System



Query Languages

Lecture No. 05



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Recap of Previous Lecture

Topic

Practice questions on R.A.



Topics to be Covered



Topic

Practice questions on R.A.

Topic

SQL

Topic

SQL clauses

Topic

Aggregate functions



★ ✓ Consider the following relational tables: ✓

✓ ★ Supplier (Sid, Sname, Rating)

✓ ★ Parts (Pid, Pname, Color)

✓ ★ Catalog (Sid, Pid, Cost)

Supplier

<u>Sid</u>	Sname	Rating
S ₁	A	3
S ₂	A	5
S ₃	B	7
S ₄	C	0

Parts

<u>Pid</u>	Pname	Color
P ₁	XYZ	Red
P ₂	PQR	Green
P ₃	XYZ	Red

Catalog

<u>Sid</u>	<u>Pid</u>	Cost
S ₁	P ₁	10
S ₂	P ₂	20
S ₃	P ₃	30

Q:- Retrieve Sids of
all the suppliers.

π_{sid} (Supplier)

o/p:

Sid
S ₁
S ₂
S ₃
S ₄

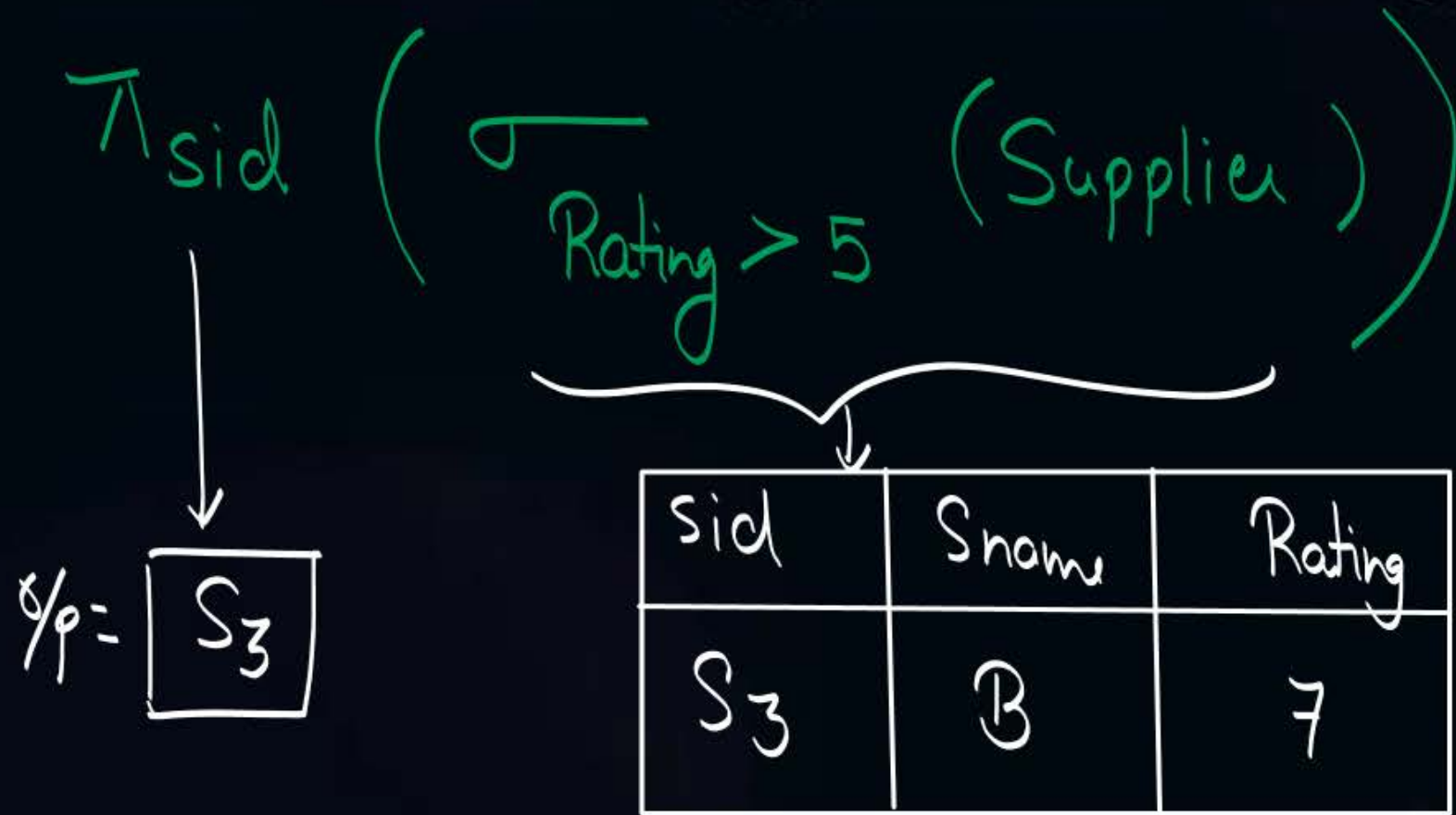
Q:- Retrieve Sids of
Suppliers who have
supplied at least one part.

π_{sid} (Catalog)

o/p =

Sid
S ₁
S ₂
S ₃

#Q. Retrieve $\checkmark$ Sid of the suppliers whose rating is more than 5.



#Q. Retrieve Sid of the suppliers who have supplied some parts.

ie. (at least one)

$\pi_{sid}(\text{Catalog}) \Rightarrow$ o/p :-

Sid
S ₁
S ₂
S ₃



#Q. Retrieve Sid of the suppliers who have supplied some Red color parts.

Catalog ✓

<u>Sid</u>	<u>Pid</u>	Cost
S ₁	P ₁	10
S ₂	P ₂	20
S ₃	P ₃	30

Pants ✓

<u>Pid</u>	Pname	Color
P ₁	XYZ	Red
P ₂	PQR	Green
P ₃	XYZ	Red

(i) $\uparrow$ Catalog.Sid

o/p:-

Sid
S ₁
S ₃

$\uparrow$ Catalog.Pid = Pants.Pid (Catalog x Pants)

$\uparrow$ Pants.Color = 'Red'

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S ₁	P ₁	10	P ₁	XYZ	Red
S ₁	P ₁	10	P ₂	PQR	Green
S ₁	P ₁	10	P ₃	XYZ	Red
S ₂	P ₂	20	P ₁	XYZ	Red
S ₂	P ₂	20	P ₂	PQR	Green
S ₂	P ₂	20	P ₃	XYZ	Red
S ₃	P ₃	30	P ₁	XYZ	Red
S ₃	P ₃	30	P ₂	PQR	Green
S ₃	P ₃	30	P ₃	XYZ	Red

Catalog

<u>Sid</u>	<u>Pid</u>	Cost
S ₁	P ₁	10
S ₂	P ₂	20
S ₃	P ₃	30

(ii) $\wedge$ Sid

↓
%p = S₁
S₃

$\left[\sigma_{\text{Color} = 'Red'} (\text{Catalog} \bowtie \text{Parts}) \right]$

Parts

<u>Pid</u>	Pname	Color
P ₁	XYZ	Red
P ₂	PQR	Green
P ₃	XYZ	Red

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S ₁	P ₁	10	P ₁	XYZ	Red
S ₁	P ₁	10	P ₂	PQR	Green
S ₁	P ₁	10	P ₃	XYZ	Red
S ₂	P ₂	20	P ₁	XYZ	Red
S ₂	P ₂	20	P ₂	PQR	Green
S ₂	P ₂	20	P ₃	XYZ	Red
S ₃	P ₃	30	P ₁	XYZ	Red
S ₃	P ₃	30	P ₂	PQR	Green
S ₃	P ₃	30	P ₃	XYZ	Red

Catalog

<u>Sid</u>	<u>Pid</u>	Cost
S ₁	P ₁	10
S ₂	P ₂	20
S ₃	P ₃	30

Parts

<u>Pid</u>	Pname	Color
P ₁	XYZ	Red
P ₂	PQR	Green
P ₃	XYZ	Red

$$(iii) \pi_{Catalog.Sid} \left(\sigma_{Catalog.Pid = Parts.Pid} \left(Catalog \times \left(\sigma_{Color = 'Red'} (Parts) \right) \right) \right)$$

$$(iv) \pi_{sid} \left(Catalog \bowtie \left(\sigma_{Color = 'Red'} (Parts) \right) \right)$$

These two query are more efficient
Compared to previous two
queries.

$$(S_C(\text{Catalog}) \times S_P(\text{Parts}))$$

$$(i) \quad \pi_{C.Sid} \left(\sigma_{\substack{C.Pid = P.Pid \\ P.Color = 'Red'}} (C \times P) \right)$$

less efficient

$$(ii) \quad \pi_{Sid} \left(\sigma_{Color='Red'} (C \bowtie P) \right)$$

$$(iii) \quad \pi_{C.Sid} \left(\sigma_{C.Pid = P.Pid} \left(C \times \left(\sigma_{Color='Red'} (P) \right) \right) \right)$$

more efficient

$$(iv) \quad \pi_{Sid} \left(C \bowtie \left(\sigma_{Color='Red'} (P) \right) \right)$$



#Q. Retrieve Sid of the suppliers who have supplied some Red or some Green color parts.

Catalog ✓

<u>Sid</u>	<u>Pid</u>	Cost
S ₁	P ₁	10
S ₂	P ₂	20
S ₃	P ₃	30

Parts ✓

<u>Pid</u>	Pname	Color
P ₁	XYZ	Red
P ₂	PQR	Green
P ₃	XYZ	Red

CXP

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S ₁	P ₁	10	P ₁	XYZ	Red
S ₁	P ₁	10	P ₂	PQR	Green
S ₁	P ₁	10	P ₃	XYZ	Red
S ₂	P ₂	20	P ₁	XYZ	Red
S ₂	P ₂	20	P ₂	PQR	Green
S ₂	P ₂	20	P ₃	XYZ	Red
S ₃	P ₃	30	P ₁	XYZ	Red
S ₃	P ₃	30	P ₂	PQR	Green
S ₃	P ₃	30	P ₃	XYZ	Red

(i)

$$\pi_{C.Sid} \left(\begin{array}{l} \sigma_{C.Pid = P.Pid} (C \times P) \\ \wedge \\ (P.Color = 'Red' \vee P.Color = 'Green') \end{array} \right)$$

(ii)

$$\pi_{Sid} \left(\sigma_{(Color = 'Red' \vee Color = 'Green')} (C \bowtie P) \right)$$

(iii)

$$\pi_{C.Sid} \left(\sigma_{C.Pid = P.Pid} \left(C \times \left(\sigma_{Color = 'Red' \vee Color = 'Green'} (P) \right) \right) \right)$$

(iv)

$$\pi_{Sid} \left(C \bowtie \left(\sigma_{Color = 'Red' \vee Color = 'Green'} (P) \right) \right)$$

#Q. Retrieve Sid of the suppliers who have supplied some Red and some Green color parts.

Catalog ✓

<u>Sid</u>	<u>Pid</u>	Cost
S ₁	P ₁	10
S ₂	P ₂	20
S ₃	P ₃	30

Parts ✓

<u>Pid</u>	Pname	Color
P ₁	XYZ	Red
P ₂	PQR	Green
P ₃	XYZ	Red

CXP

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
→ S ₁	P ₁	10	P ₁	XYZ	Red
S ₁	P ₁	10	P ₂	PQR	Green
S ₁	P ₁	10	P ₃	XYZ	Red
S ₂	P ₂	20	P ₁	XYZ	Red
→ S ₂	P ₂	20	P ₂	PQR	Green
S ₂	P ₂	20	P ₃	XYZ	Red
S ₃	P ₃	30	P ₁	XYZ	Red
S ₃	P ₃	30	P ₂	PQR	Green
→ S ₃	P ₃	30	P ₃	XYZ	Red

(i)

$$\pi_{C.Sid} \left(\sigma_{C.Pid = P.Pid \wedge (P.Color = 'Red' \wedge P.Color = 'Green')} (C \times P) \right)$$

(ii)

$$\pi_{Sid} \left(\sigma_{(Color = 'Red' \wedge Color = 'Green')} (C \bowtie P) \right)$$

$$(iii) \pi_{C.Sid} \left(\sigma_{C.Pid = P.Pid} \left(C \times \left(\sigma_{Color = 'Red' \wedge Color = 'Green'} (P) \right) \right) \right)$$

$$(iv) \pi_{Sid} \left(C \bowtie \left(\sigma_{Color = 'Red' \wedge Color = 'Green'} (P) \right) \right)$$

All are

Wrong query
to get the
desired result.

⇓

In a single tuple
of "C x P", Color
Can never be "Red"
as well as "green" at
the same time

∴ All
of them will
Produce Empty output

$$(i) \quad \pi_{C.Sid} \left(\sigma_{\substack{C.Pid = P.Pid \\ P.Color = 'Red'}} (C \times P) \right) \cap \pi_{\underline{C}.Sid} \left(\sigma_{\substack{\underline{C}.Pid = \underline{P}.Pid \\ \underline{P}.Color = 'Green'}} (\underline{C} \times \underline{P}) \right)$$

$$(ii) \quad \pi_{Sid} \left(\sigma_{Color = 'Red'} (C \bowtie P) \right) \cap \pi_{Sid} \left(\sigma_{Color = 'Green'} (\underline{C} \bowtie \underline{P}) \right)$$

$$(iii) \quad \pi_{C.Sid} \left(\sigma_{C.Pid = P.Pid} \left(C \times \left(\sigma_{Color = 'Red'} (P) \right) \right) \right) \cap \pi_{\underline{C}.Sid} \left(\sigma_{\underline{C}.Pid = \underline{P}.Pid} \left(\underline{C} \times \left(\sigma_{Color = 'Green'} (\underline{P}) \right) \right) \right)$$

$$(iv) \quad \pi_{Sid} \left(C \bowtie \left(\sigma_{Color = 'Red'} (P) \right) \right) \cap \pi_{Sid} \left(\underline{C} \bowtie \left(\sigma_{Color = 'Green'} (\underline{P}) \right) \right)$$

Catalog ✓

<u>Sid</u>	<u>Pid</u>	Cost
S ₁	P ₁	10
S ₂	P ₂	20
S ₃	P ₃	30

Parts ✓

<u>Pid</u>	<u>Pname</u>	<u>Color</u>
P ₁	XYZ	Red
P ₂	PQR	Green
P ₃	XYZ	Red

C₁ × P₁

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S ₁	P ₁	10	P ₁	XYZ	Red
S ₁	P ₁	10	P ₂	PQR	Green
S ₁	P ₁	10	P ₃	XYZ	Red
S ₂	P ₂	20	P ₁	XYZ	Red
S ₂	P ₂	20	P ₂	PQR	Green
S ₂	P ₂	20	P ₃	XYZ	Red
S ₃	P ₃	30	P ₁	XYZ	Red
S ₃	P ₃	30	P ₂	PQR	Green
S ₃	P ₃	30	P ₃	XYZ	Red

C₂ × P₂

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S ₁	P ₁	10	P ₁	XYZ	Red
S ₁	P ₁	10	P ₂	PQR	Green
S ₁	P ₁	10	P ₃	XYZ	Red
S ₂	P ₂	20	P ₁	XYZ	Red
S ₂	P ₂	20	P ₂	PQR	Green
S ₂	P ₂	20	P ₃	XYZ	Red
S ₃	P ₃	30	P ₁	XYZ	Red
S ₃	P ₃	30	P ₂	PQR	Green
S ₃	P ₃	30	P ₃	XYZ	Red

$$= \pi_{C_1.Sid} \left(\sigma_{\left((C_1.Sid = C_2.Sid) \wedge (C_1.Pid = P_1.Pid \wedge P_1.Color = 'Red') \wedge (C_2.Pid = P_2.Pid \wedge P_2.Color = 'green') \right)} \left((C_1 \times P_1) \times (C_2 \times P_2) \right) \right)$$

let us use $C_1 \times P_1$ to select the tuple w.r.t. 'Red' Color part
and
let us use $C_2 \times P_2$ to select the tuple w.r.t. 'green' Color part

we can also use $C_2.Sid$ as well, because both of them are same

Q:- Retrieve Sids of suppliers who have supplied some "red" or some "green" color parts

$$(i) \pi_{C.Sid} \left(\sigma_{C.Pid = P.Pid \wedge P.Color = 'Red'} (C \times P) \right) \cup \pi_{\underline{C}.Sid} \left(\sigma_{\underline{C}.Pid = \underline{P}.Pid \wedge \underline{P}.Color = 'Green'} (\underline{C} \times \underline{P}) \right)$$

$$(ii) \pi_{Sid} \left(\sigma_{Color = 'Red'} (C \bowtie P) \right) \cup \pi_{Sid} \left(\sigma_{Color = 'Green'} (\underline{C} \bowtie \underline{P}) \right)$$

$$(iii) \pi_{C.Sid} \left(\sigma_{C.Pid = P.Pid} \left(C \times \left(\sigma_{Color = 'Red'} (P) \right) \right) \right) \cup \pi_{\underline{C}.Sid} \left(\sigma_{\underline{C}.Pid = \underline{P}.Pid} \left(\underline{C} \times \left(\sigma_{Color = 'Green'} (\underline{P}) \right) \right) \right)$$

$$(iv) \pi_{Sid} \left(C \bowtie \left(\sigma_{Color = 'Red'} (P) \right) \right) \cup \pi_{Sid} \left(\underline{C} \bowtie \left(\sigma_{Color = 'Green'} (\underline{P}) \right) \right)$$

Q: Retrieve Sids of the suppliers who have supplied some "red" or some "green" color parts

$$= \pi_{C_1.Sid} \left(\sigma_{\left((C_1.Sid = C_2.Sid) \wedge \left((C_1.Pid = P_1.Pid \wedge P_1.Color = 'Red') \vee (C_2.Pid = P_2.Pid \wedge P_2.Color = 'green') \right) \right)} \left((C_1 \times P_1) \times (C_2 \times P_2) \right) \right)$$

let us use $C_1 \times P_1$ to select the tuple w.r.t. 'Red' color part

and let us use $C_2 \times P_2$ to select the tuple w.r.t. 'green' color part

we can also use $C_2.Sid$ as well, because both of them are same

#Q. Retrieve Sid of the suppliers who have supplied all parts.

Catalog (Sid, Pid, Cost)

Parts (Pid, Pname, Color)

We are interested in the
Sids in the catalog table
which are associated
with all Pids of Parts table

so query will be

$$\pi_{\text{Sid, Pid}}(\text{Catalog}) \div \pi_{\text{Pid}}(\text{Parts})$$

$$= \left[\pi_{\text{Sid}}(C) - \pi_{\text{Sid}} \left[\left(\pi_{\text{Sid}}(C) \times \pi_{\text{Pid}}(P) \right) - \pi_{\text{Sid, Pid}}(C) \right] \right]$$

#Q. Retrieve Sid of the suppliers who have supplied all Red color parts.

We are actually looking for Sids in Catalog table which are associated with all Pids of "Red" Color Parts.

$$\pi_{\text{Sid}, \text{Pid}}(C) \div \pi_{\text{Pid}}(\sigma_{\text{Color}='Red'}(P))$$

$$= \left[\pi_{\text{Sid}}(C) - \pi_{\text{Sid}} \left[\left(\pi_{\text{Sid}}(C) \times \pi_{\text{Pid}}(\sigma_{\text{Color}='Red'}(P)) \right) - \pi_{\text{Sid}, \text{Pid}}(C) \right] \right]$$

#Q. Retrieve Sid of the suppliers who have supplied at least two parts.

* For the upcoming questions we will consider the following Catalog relation.

Catalog

<u>Sid</u>	<u>Pid</u>	Cost
S ₁	P ₁	20
S ₁	P ₂	30
S ₂	P ₂	30
S ₃	P ₂	20
S ₃	P ₃	10

Q. Retrieve Sids of the Suppliers who have supplied at least two parts.

Catalog (C ₁)			Catalog (C ₂)		
Sid	Pid	Cost	Sid	Pid	Cost
S ₁	P ₁	20	S ₁	P ₁	20
S ₁	P ₂	30	S ₁	P ₂	30
S ₂	P ₂	30	S ₂	P ₂	30
S ₃	P ₂	20	S ₃	P ₂	20
S ₃	P ₃	10	S ₃	P ₃	10

We will have to compare the Sid of a tuple of Catalog table with Sids of all other tuples of the Catalog table.

If the same Sid is present in any other tuple of Catalog table, that Supplier must have supplied at least two parts.

$$= \pi_{C_1.Sid} \left[\sigma_{((C_1.Sid = C_2.Sid) \wedge (C_1.Pid \neq C_2.Pid))} (C_1 \times C_2) \right]$$

If we do not use this Condⁿ, then each Sid from Catalog will be present in o/p

Because each tuple will combine with itself

#Q. Retrieve Sid of the suppliers who have supplied at most one part.

ie., "0 or 1"

= All suppliers — Suppliers who have supplied at least two parts
 $\{0 \text{ or more parts}\} - \{2 \text{ or more parts}\}$

= $\pi_{\text{sid}}(\text{Supplier}) - \pi_{c_1.\text{sid}} \left(\sigma_{\substack{(c_1.\text{sid} = c_2.\text{sid}) \\ (c_1.\text{pid} \neq c_2.\text{pid})}} (c_1 \times c_2) \right)$

* H.W. ✓



#Q. Retrieve Sid of the suppliers who have supplied at least three parts.

$$(C_1 \times C_2 \times C_3)$$

Catalog

<u>Sid</u>	<u>Pid</u>	Cost
S ₁	P ₁	20
S ₁	P ₂	30
S ₂	P ₂	30
S ₃	P ₂	20
S ₃	P ₃	10
S ₄	P ₃	20

$\bigwedge_{C_1.Sid} \left[\begin{aligned} & \left((C_1.Sid = C_2.Sid) \wedge (C_2.Sid = C_3.Sid) \wedge (C_1.Sid = C_3.Sid) \right) (C_1 \times C_2 \times C_3) \\ & \wedge \\ & \left((C_1.Pid \neq C_2.Pid) \wedge (C_2.Pid \neq C_3.Pid) \wedge (C_1.Pid \neq C_3.Pid) \right) \end{aligned} \right]$

it is optional we can skip this

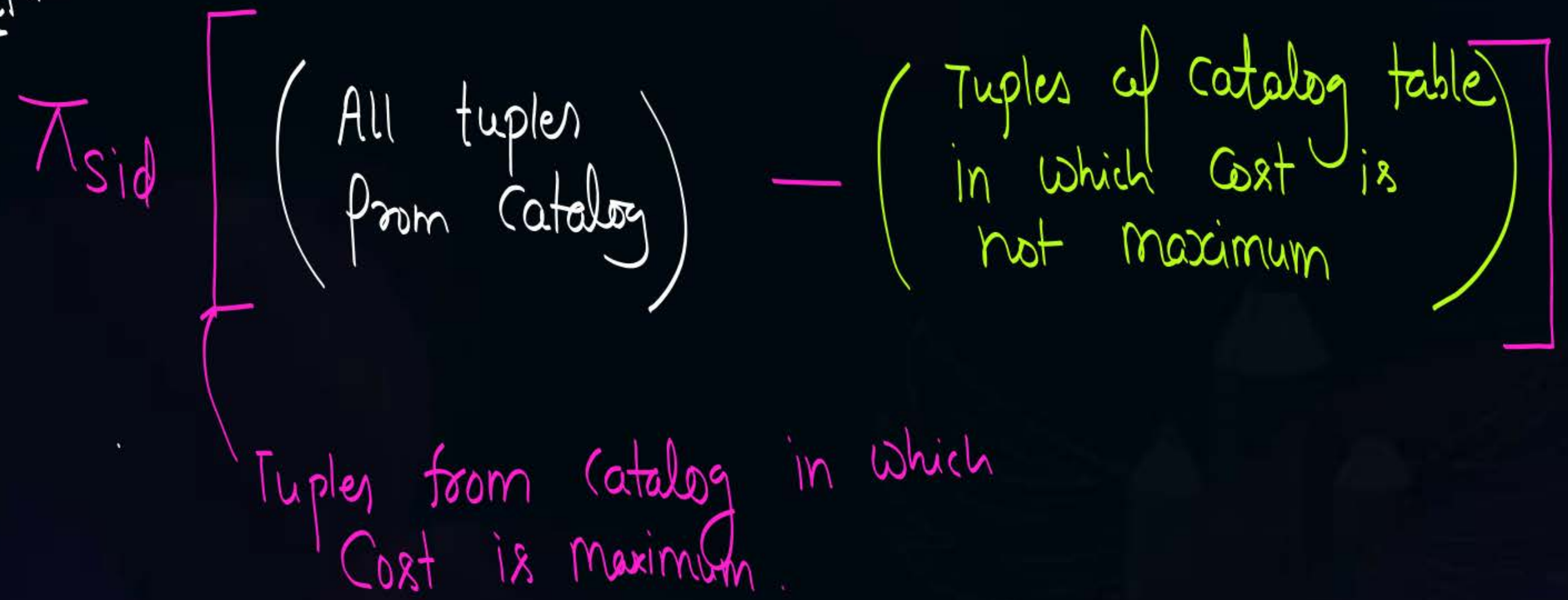
when we are checking w.r.t "not equal" this condⁿ is mandatory

H.W.



#Q. Retrieve Sids of the suppliers who have supplied most expensive parts.

Hint?



Query: - Retrieve Sids of the suppliers who have supplied most expensive parts.

Catalog (C₂)

Sid	Pid	Cost
S ₁	P ₁	20
S ₁	P ₂	30
S ₂	P ₂	30
S ₃	P ₂	20
S ₃	P ₃	10

Catalog (C₂)

Sid	Pid	Cost
S ₁	P ₁	20
S ₁	P ₂	30
S ₂	P ₂	30
S ₃	P ₂	20
S ₃	P ₃	10

output of this subtraction opn will be the tuples from Catalog in which cost is maximum

π_{Sid}
o/p:-

Sid
S ₁
S ₂

Sid	Pid	Cost
S ₁	P ₁	20
S ₁	P ₂	30
S ₂	P ₂	30
S ₃	P ₂	20
S ₃	P ₃	10

Sid	Pid	Cost
S ₁	P ₂	30
S ₂	P ₂	30

This query will produce the tuples from Catalog table, st. the Cost is not maximum in those tuples

$\pi_{C_1.Sid, C_1.Pid, C_1.Cost} \left(\sigma_{C_1.Cost < C_2.Cost} (C_1 \times C_2) \right)$

Sid	Pid	Cost
S ₁	P ₁	20
S ₃	P ₂	20
S ₃	P ₃	10

Query: - Retrieve Sids of the suppliers who have supplied most Expensive Parts.

Catalog (C₂)

Sid	Pid	Cost
S ₁	P ₁	20
S ₁	P ₂	30
S ₂	P ₂	30
S ₃	P ₂	20
S ₃	P ₃	10

Catalog (C₂)

Sid	Pid	Cost
S ₁	P ₁	20
S ₁	P ₂	30
S ₂	P ₂	30
S ₃	P ₂	20
S ₃	P ₃	10

This query will produce the tuples from Catalog table, st. the Cost is not maximum in those tuples

$$= \pi_{\text{Sid}}(\text{Catalog}) - \pi_{C_1.\text{Sid}} \left(\pi_{C_1.\text{Sid}, C_1.\text{Pid}, C_1.\text{Cost}} \left(\sigma_{C_1.\text{Cost} < C_2.\text{Cost}} (C_1 \times C_2) \right) \right)$$

o/p of this query will not be correct, if the supplier who have supplied a part at maximum Cost has also supplied some other parts at lower Cost

Query: - Retrieve Sids of the suppliers who have supplied least expensive parts.

Catalog (C₂)

Sid	Pid	Cost
S ₁	P ₁	20
S ₁	P ₂	30
S ₂	P ₂	30
S ₃	P ₂	20
S ₃	P ₃	10

Catalog (C₂)

Sid	Pid	Cost
S ₁	P ₁	20
S ₁	P ₂	30
S ₂	P ₂	30
S ₃	P ₂	20
S ₃	P ₃	10

output of this subtraction opn will be the tuples from Catalog in which cost is minimum

π_{Sid}
 $\rho =$

Sid
S ₃

Sid	Pid	Cost
S ₁	P ₁	20
S ₁	P ₂	30
S ₂	P ₂	30
S ₃	P ₂	20
S ₃	P ₃	10

Sid	Pid	Cost
S ₃	P ₃	10

This query will produce the tuples from Catalog table, s.t. the Cost is not minimum in those tuples

$\pi_{C_1.Sid, C_1.Pid, C_1.Cost} \left(\sigma_{C_1.Cost > C_2.Cost} (C_1 \times C_2) \right)$

Sid	Pid	Cost
S ₁	P ₁	20
S ₁	P ₂	30
S ₂	P ₂	30
S ₃	P ₂	20



Topic : SQL

Prerequisite :- Relational Algebra



→ Structured Query language

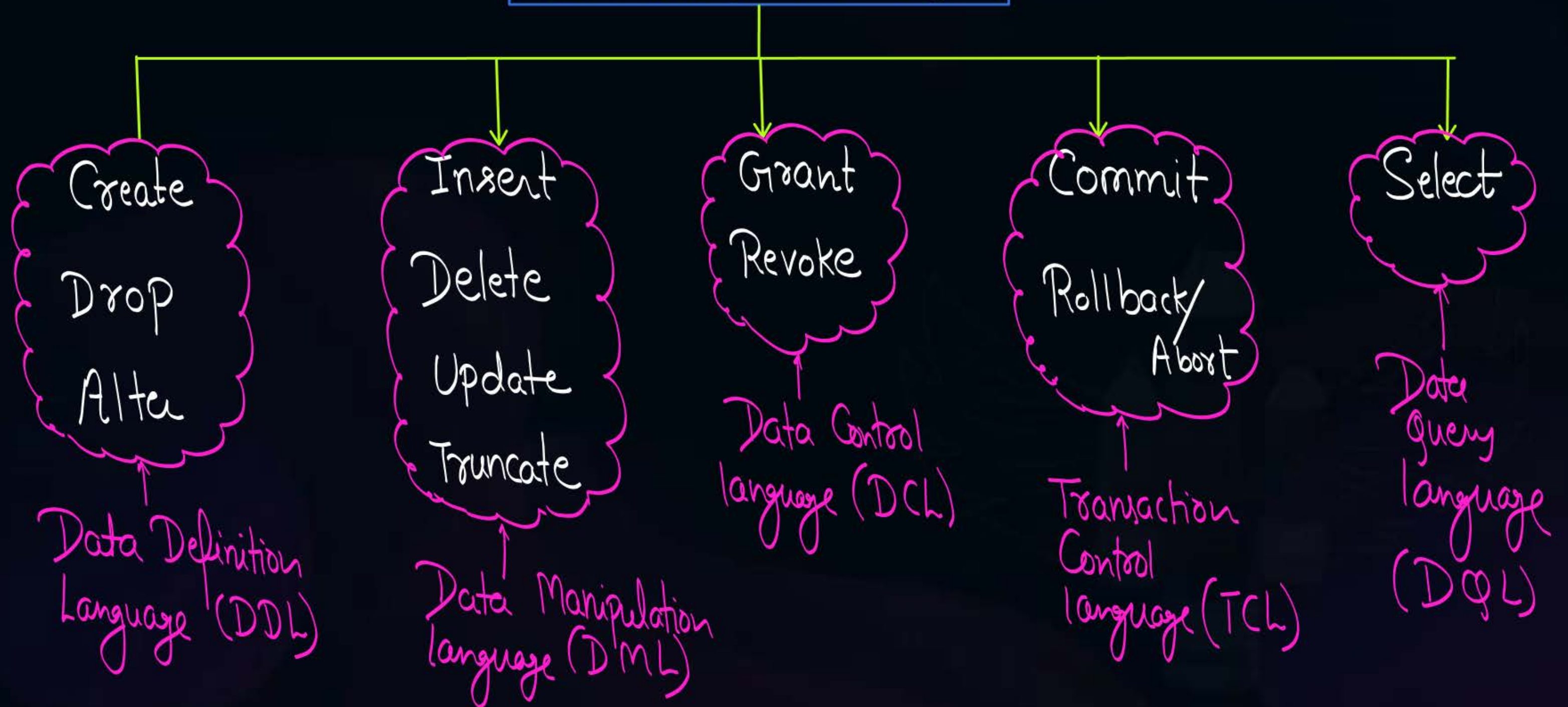
→ It is a non-procedural query language

∴ We must
Understand the
"Syntax" of SQL



Topic : Commands in SQL

Commands in SQL





Topic : SQL clauses

→ Some basic SQL Clauses are :-

- ① From
- ② Where
- ③ Group By
- ④ Having
- ⑤ Order By



Topic : SQL clauses



$\uparrow A_1, A_2, \dots, A_K \equiv$

① Select

[distinct]

$A_1, A_2, \dots, A_K$

it is optional.

If we do not use "distinct" then o/p of SQL query may contain duplicate tuples

List of attributes required in o/p

$R_1 \times R_2 \times \dots \times R_m \equiv$

② From $R_1, R_2, \dots, R_m$

$\int_{(Cond^n \text{ to select tuples})} \equiv$

③ Where (Condⁿ to select tuples)

List of relations required for the execution of query

No Equivalent R.A. Expression

④ Group by (Attributes)

⑤ Having (Condⁿ)

⑥ Order by (Attribute) DESC/ASC

Condⁿ specified with "having" clause will be applied on "each group, provided group by clause is present"

Select $\not\equiv$ Projection (π)
(in SQL) (in Relational Algebra)

Select distinct $\equiv$ Projection (π)
(in SQL) (in Relational Algebra)



Topic : Aggregate functions

* There are five aggregate functions available in SQL

① Count

② Sum

③ Avg

④ Max

⑤ Min



Topic : Aggregate functions



H.W.

- * Select Count (*)
from STUDENT = ?
- * Select Count (Marks)
from STUDENT = ?
- * Select Count (distinct Marks)
from STUDENT = ?

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions



Ex

→ Select Sum(Marks)
from STUDENT = ?

→ Select Sum(distinct Marks)
from STUDENT = ?

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions



H.W.

Select Avg(Marks)
From STUDENT = ?

Select Avg (distinct Marks)
From STUDENT = ?

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions



H.W.

- Select Max(Marks)
from STUDENT = ?
- Select Max(distinct Marks)
from STUDENT = ?

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions



H.W.

- Select Min (Marks)
from STUDENT = ?
- Select Min (distinct Marks)
from STUDENT = ?

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



2 mins Summary



✓ **Topic** Practice questions on R.A.

✓ **Topic** SQL

✓ **Topic** SQL clauses

✓ **Topic** Aggregate functions

THANK - YOU